

HCI- LABSHEET 2

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Activity 1

Source	Participant	Demographic	Topic	Key Finding	Quote
Interview	User 01	Male, 32, Busy Professional	Workout Plans	Plans are too rigid	"I can't follow the same workout every day."
Interview	User 02	Female, 27, Beginner	Motivation	Feels alone during workouts	"I wish I had friends keeping me accountable."
Survey	420 Users	Mixed	Onboarding	68% left after onboarding	"Setup takes too long."
Diary Study	User 11	Female, 24	Nutrition	Logging meals is repetitive	"Entering every ingredient is exhausting."
Firebase Analytics	All Users	Mixed	Retention	Users stop after first week	High drop-off after Day 3
App Store Reviews	Multiple	Mixed	Overall Experience	App feels outdated	"Needs smarter recommendations."

Organize by

- Interviews
- Surveys
- Diary Studies
- Observation Notes
- Analytics
- App Store Reviews

Initial Topics

- Onboarding
- Workout Experience
- Nutrition Tracking
- Motivation

- Social Features

Activity 2

Step 1: Coding

Quote	Initial Code
"Workout plans never fit my schedule."	Generic workouts
"Logging food is annoying."	Difficult nutrition logging
"Nobody keeps me motivated."	Lack of accountability
"The app doesn't know me."	No personalization
"I stopped after three days."	Poor onboarding

Step 2: Themes

Theme	Supporting Codes	Evidence
Lack of Personalization	Generic workouts, Fixed schedules	Mentioned by 25 interviewees
Nutrition Tracking Friction	Manual logging, Time consuming	Mentioned in diary studies and surveys
Poor Motivation	No rewards, No accountability	Mentioned in focus groups
Social Isolation	No community, Workout alone	Mentioned in interviews
Poor Onboarding	Complex setup	Analytics show 68% drop-off

Theme Summary

Theme 1: Lack of Personalization

Users want AI recommendations that adapt to their schedules.

Theme 2: Poor Motivation

Users lose interest because there are no rewards or encouragement.

Theme 3: Nutrition Tracking

Meal logging takes too much time.

Theme 4: Social Isolation

Users want workout partners and challenges.

Theme 5: Onboarding Problems

Most users abandon the app during the first experience.

Activity 3

PERSONALIZATION

- Workout plans are repetitive
- Needs AI suggestions
- Plans don't fit schedules
- No adaptive workouts

MOTIVATION

- No achievements
- No reminders
- No celebrations
- Lose interest quickly

NUTRITION TRACKING

- Typing meals takes too long
- Too many steps
- Needs camera scanning

SOCIAL FEATURES

- No friends
- Need workout groups
- Want community challenges

ONBOARDING

- Too many setup screens
- Confusing first use

Key Findings

- Personalization is the biggest issue.
- Nutrition logging creates high friction.
- Social interaction is missing.
- Onboarding needs simplification.

Activity 4

Persona 1 – Alex Rivera

Age: 32

Occupation: Marketing Executive

Fitness Level: Intermediate

Goals

- Stay fit despite busy work
- Short workouts

Frustrations

- Workout plans never adapt
- Doesn't have much time

Empathy Map

Says : "I only have 20 minutes today."

Thinks: "I wish the app understood my schedule."

Does

Exercises 3–4 times weekly.

Feels

Frustrated when workouts are too long.

Pains

- Rigid plans

- Time pressure

Gains

- AI adaptive workouts
- Flexible scheduling

Persona 2 – Priya Singh

Age: 27

Occupation: Teacher

Fitness Level: Beginner

Goals

- Lose weight
- Stay motivated

Frustrations

- Exercises alone
- Doesn't know where to start

Empathy Map

Says: "I wish someone encouraged me."

Thinks: "I'm afraid I'll quit."

Does

Walks occasionally.

Feels

Lonely.

Pains

- No accountability
- Beginner uncertainty

Gains

- Community groups

- Challenges

Persona 3 – Michael Chen

Age: 35

Occupation: Software Engineer

Fitness Goal

Healthy eating

Frustrations

- Food logging takes forever

Empathy Map

Says: "I don't want to type every meal."

Thinks: "There must be an easier way."

Does

Exercises regularly.

Feels

Annoyed.

Pains

- Manual meal entry

Gains

- Camera-based nutrition logging

Activity 5

Priority	User Story
High	As a busy professional, I want AI-generated workout plans so that I can exercise despite my changing schedule.
High	As a beginner, I want beginner-friendly workout guidance so that I feel confident exercising.
High	As a health-conscious user, I want camera-based food recognition so that meal logging is faster.
High	As a user, I want personalized workout recommendations so that my fitness plan matches my goals.
Medium	As a user, I want to join fitness communities so that I stay motivated.
Medium	As a user, I want progress celebrations so that I remain engaged.
Medium	As a user, I want daily reminders so that I don't miss workouts.
Medium	As a user, I want private workout groups so that I can exercise with friends.
Low	As a user, I want offline workout access so that I can use the app anywhere.
Low	As a user, I want detailed progress dashboards so that I can track improvements.

Activity 6

Current State

Stage	Action	Pain Point
Discover	Downloads app	None
Onboarding	Creates account	Too many steps
Workout	Starts plan	Generic workouts
Nutrition	Logs meals	Manual entry
Social	Looks for friends	No community
Long-term	Stops using app	Loses motivation

Future State

Stage	Action	Opportunity
Discover	Downloads app	Clear benefits
Onboarding	Quick setup	AI customizes immediately
Workout	Adaptive AI plan	Personalized workouts
Nutrition	Camera scan	Faster logging
Social	Joins challenge	Community support
Long-term	Earns achievements	Higher engagement

Activity 7

1. How might we create AI-powered workout plans that adapt to each user's changing schedule and fitness goals?
2. How might we encourage users to stay motivated through meaningful social connections and community challenges?
3. How might we make nutrition tracking fast, accurate, and effortless using camera-based food recognition?

Activity 8

Functional Requirements

ID	Requirement	Priority
FR1	System shall generate AI-personalized workout plans.	High
FR2	System shall allow users to join private fitness groups.	High
FR3	System shall support camera-based nutrition logging.	High
FR4	System shall display progress dashboards.	Medium
FR5	System shall send workout reminders.	Medium
FR6	System shall recommend community challenges.	Medium

Non-Functional Requirements

ID	Requirement	Priority
NFR1	App response time shall be under 2 seconds.	High
NFR2	System shall comply with GDPR and CCPA.	High
NFR3	App shall support offline workout access.	Medium
NFR4	Interface shall meet accessibility guidelines	High
NFR5	System shall encrypt all user data.	High

Traceability Matrix

Requirement	Theme	Persona	User Story	HMW
FR1	Personalization	Alex	AI workout plan	HMW 1
FR2	Social Isolation	Priya	Join community	HMW 2
FR3	Nutrition Tracking	Michael	Camera logging	HMW 3
FR4	Motivation	Alex	Progress dashboard	HMW 2
FR5	Motivation	Priya	Workout reminders	HMW 2
NFR1	Performance	All	Smooth experience	Supports all
NFR2	Privacy	All	Secure app	Supports all
NFR3	Accessibility	All	Offline access	Supports all